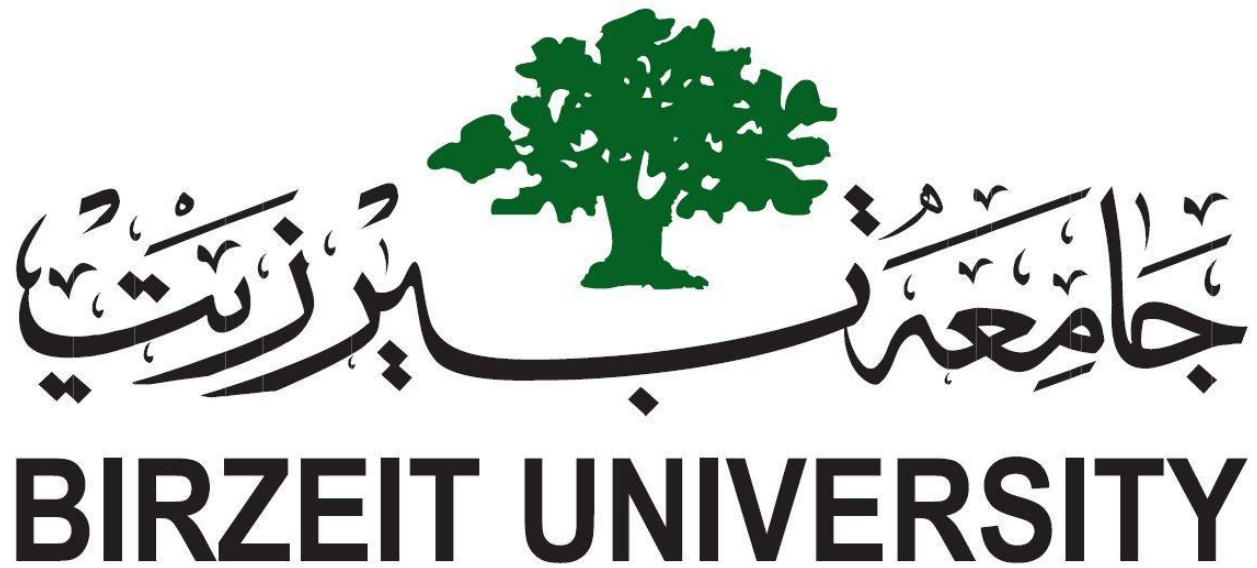




**Data Science**

Project Phase 2

**Flight Delay Prediction and Airport Punctuality Analysis Across  
Five European Countries**

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## Abstract

Flight delays are a common challenge in air transportation, affecting passenger experience, airline operations, airport efficiency, and the reliability of connected flight networks [1]. This project, titled “Flight Delay Prediction and Airport Punctuality Analysis Across Five European Countries,” investigates airport punctuality and delay patterns across Ireland, Denmark, Portugal, Belgium, and Austria. The project aims to build a complete data science pipeline that includes data collection, cleaning, preprocessing, exploratory analysis, visualization, and a simple machine learning model for predicting a delay-related target.

The analysis focuses on airport-level, geographic, time-based, and operational features in order to compare punctuality performance across the selected countries and identify patterns that may contribute to flight delays. Since airports and air traffic operations depend on many connected systems working together, studying delay behavior can provide useful insights into airport performance and operational reliability [2][3]. The expected outcome is a clear project that presents the results, explains what they mean, and discusses the limits of the data and the model. By using real aviation data, this project can also help future data science students understand how data science can be applied to flight delay prediction and airport punctuality analysis.

## 1. Introduction

Flight delays are a major issue for everyone involved in the aviation system. Airlines lose money because of extra fuel, crew costs, and disrupted scheduling, while passengers lose time, miss connections, and experience reduced travel quality. When one flight is delayed, the impact often spreads through the network and affects many other flights, sometimes across different countries. Because of this domino effect, predicting delays accurately is an important problem in modern data science.

This project builds a system to study and predict flight delays across five European countries: Ireland, Portugal, Belgium, Denmark, and Austria. To do this, we bring together data on flights, airports, and weather in order to understand the main reasons behind delays. The goal is to design a clear pipeline that takes raw data, prepares it through

cleaning and feature engineering, and produces a machine learning model that can tell us whether a flight is likely to be on time or delayed.

The report is organized as follows. Section 2 (Background) explains what flight delays are and why they happen. Section 3 (Domain Explanation) describes airport punctuality, the data we are working with, and the reasons for selecting the five chosen countries. Section 4 (Algorithm Background) introduces the main machine learning algorithms relevant to this project. Section 5 reviews five recent research papers that are closely related to our work and discusses how each of them informs our approach. Finally, Section 6 lists the references used throughout this document.

## 2. Background

Flight delays happen when a flight does not depart or arrive at its planned time. This is a common problem in air transportation and it affects passengers, airlines, and airports together. Passengers may miss other flights or wait for long periods, while airlines may lose money because of extra fuel use, crew scheduling problems, and operational disruptions.

There are many reasons for flight delays. Weather is one of the main causes, including fog, rain, snow, strong wind, or low visibility. Airport traffic can also cause delays when many flights arrive or depart at the same time. Other reasons include technical problems with the aircraft, late arrival of the inbound flight, air traffic control restrictions, and boarding issues at the gate.

Flight delays can also be affected by the country, airport location, season, and flight schedule. For example, summer often has more flights because of holiday travel, while winter usually has more weather-related problems. Because of these many factors, predicting delays is not a simple task. Machine learning is a useful approach because it can learn from previous flight data, find patterns, and help predict whether a flight may be delayed. This kind of prediction can support better planning for airports and airlines, and can also give passengers more reliable information.

### 3. Domain Explanation

Airport punctuality describes how well flights follow their scheduled times. If most flights at an airport arrive and depart on time, that airport is considered more punctual. Punctuality is important because it shows how reliable airport operations are and how well the airport handles its daily traffic.

Delay data usually includes the scheduled time, the actual time, the delay duration, the airport name, the country, the airline, and the date. This kind of data allows us to compare planned flight times with real flight performance and to study which factors are most strongly associated with delays.

In this project, we focus on five European countries: Ireland, Portugal, Belgium, Denmark, and Austria. These countries were chosen because they belong to different parts of Europe and may have different travel patterns, weather conditions, and levels of airport traffic. Studying these countries allows us to compare airport punctuality across different settings and to understand how factors such as country, season, and airport traffic can affect flight delays. This domain is useful because predicting delays can help airports and airlines improve their planning and provide passengers with better information.

### 4. Algorithm Background

Several machine learning algorithms are commonly used for flight delay prediction. In this project, we focus on three algorithms that are well suited for tabular data and that allow us to compare different modeling approaches.

#### 4.1 Decision Tree

A Decision Tree is a visual and logical algorithm that works like a flowchart of yes/no questions. For example, it may first ask: "Is the wind speed high?" If the answer is yes, it moves to the next question, such as: "Is it a weekend?" By following these branches, the model reaches a final decision about whether the flight is likely to be delayed. Decision Trees are a strong baseline because they are easy to explain and we can clearly see how the model makes its decisions, which is useful for an educational project.

#### 4.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is based on the idea of similarity. To classify a new flight, the model looks at the most similar flights in the training data (the "neighbors") and checks how they behaved. If most of these similar flights were delayed, the model predicts that the new

flight will also be delayed. KNN is intuitive and works well when flights with similar conditions, such as the same airport and time of day, tend to behave in similar ways.

### 4.3 XGBoost

XGBoost is one of the most powerful and widely used algorithms in current data science practice. It is an advanced version of decision trees that builds many small trees one after another, where each new tree tries to fix the mistakes of the previous ones. It is fast, scalable, and able to handle our combined dataset across five countries. Importantly, XGBoost also reports feature importance, which helps us see which factors, such as wind speed, airline, or scheduled time, are the most important reasons for delays.

## 5. Review of Related Papers

In this section, we review five recent research papers that are closely related to our project. For each paper, we summarize the main idea and discuss how it connects to our work on flight delay prediction and airport punctuality analysis.

### 5.1 Paper 1 — Probabilistic Flight Delay Predictions Using Machine Learning and Applications to the Flight-to-Gate Assignment Problem

#### Paper Review

This paper studies flight delay prediction using machine learning, but it focuses on predicting the probability of delay rather than only predicting one exact delay value or a simple delayed/not-delayed result. The authors explain that, in airport operations, knowing the uncertainty of a delay can be very useful because airport planners need to prepare for many possible situations. The paper uses two machine learning methods: Mixture Density Networks and Random Forest Regression. These methods are used to predict arrival and departure delay distributions based on flight schedule data and weather data.

The results showed that both methods were able to provide useful delay predictions, with Random Forest Regression performing slightly better in some cases. The paper also connects delay prediction to the flight-to-gate assignment problem, where airports need to assign aircraft to gates while avoiding conflicts. By using probabilistic delay predictions, the model reduced aircraft gate conflicts by up to 74% compared to a normal deterministic gate assignment model. This shows that flight delay prediction is not only useful for knowing whether a flight may be late, but it can also help improve airport planning and operations [4].

#### Connection to Our Project

This paper is useful for our project because it shows how flight delay prediction can be connected to real airport performance. Our project focuses on airport punctuality and delay analysis across Ireland, Denmark, Portugal, Belgium, and Austria, and this paper supports the idea that delay prediction can help explain and improve airport operations.

It also shows the importance of using different types of features, such as time-based, airport-related, and weather-related features. Although our project will use simpler methods, the paper gives a clear example of how machine learning can be applied to flight delay data in a practical way. It also reminds us that the goal is not only to build a model, but also to explain how the results can be useful in real aviation situations [4].

## 5.2 Paper 2 — Spatiotemporal Propagation Learning for Network-Wide Flight Delay Prediction

### Paper Review

This paper focuses on predicting flight delays across a whole airport network. The main idea is that delays can spread from one airport to another because airports are connected through flights. For example, a delay at one airport can affect other airports later in the day, especially when many flights travel between them. The paper explains that flight delay prediction should consider both spatial factors, such as the relationships between airports, and temporal factors, such as how delays change over time.

The authors propose a model called the SpatioTemporal Propagation Network (STPN). This model uses ideas from graph neural networks and self-attention. In simple terms, the model treats airports as connected points in a network and tries to learn how delays move between them over time. The paper also considers factors such as airport distance, flight schedule connections, traffic volume, and weather conditions. The results showed that STPN performed better than several other models in predicting both arrival and departure delays, especially for longer prediction horizons. This demonstrates that understanding the relationships between airports is important when studying flight delays [5].

### Connection to Our Project

This paper is useful for our project because our topic also studies airport punctuality and flight delay patterns across multiple countries. Since our project covers Ireland, Denmark, Portugal, Belgium, and Austria, it is important to remember that airports do not work as isolated places. Delay patterns can be affected by time, airport location, flight connections, and weather. Even though our project uses simpler methods, this paper gives us a strong idea that delay prediction becomes more meaningful when we look at airports as part of a connected system.



The paper also supports the importance of using different types of features in our dataset. For example, we can benefit from time-based features, airport information, country information, and possibly weather-related features. It also reminds us that the goal is not only to predict delays, but also to explain why some airports or some periods may have worse punctuality than others. This matches our project goal of building a clear analysis that students can read and understand [5].

### 5.3 Paper 3 — From Data to Complex Network Control of Airline Flight Delays

#### Paper Review

This research takes a network-based approach to the aviation system rather than looking at flights in isolation. The authors build a network in which airports are nodes and the routes between them are links. They then use a technique called network control to study how a single delay can propagate like a wave across many cities and countries. This helps the reader understand the “ripple effect” of global air travel and how different airports influence each other through their shared schedules [6].

One of the most interesting findings is that reducing delays at major hub airports can lower total system delays by around 30%. This shows that not all airports contribute equally to overall delays: a delay at a large hub has a much bigger impact than a delay at a small airport, because the hub serves as a connection point for many other flights [6].

#### Connection to Our Project

This paper inspires us to include features that describe airport size and airport type in our dataset. Since our project covers five countries with airports of different sizes, distinguishing between large hubs and smaller regional airports may help our model become more accurate. The paper also reinforces the idea that delays at certain airports are particularly important to predict and explain, because they can influence many other flights in the network [6].

### 5.4 Paper 4 — Prediction of Air Traffic Delays: An Agent-Based Model

#### Paper Review

In this paper, the authors use a method called Agent-Based Modeling, which is a type of simulation. They create digital agents to represent planes, airports, and ground crews, and they observe how these agents interact in real time. This approach is interesting

because it tries to mimic real-life situations, showing how a late arrival can cause the ground crew to be late, which then causes the next flight to depart late as well [7].

A key finding of the study is that buffer time, which is the gap between scheduled flights, is one of the most critical factors that influence whether a delay will happen. Short buffer times leave little room for recovery, so even a small disruption can cause a chain of delays [7].

### Connection to Our Project

This paper directly motivates us to create time-based features in our dataset, such as the time of day, the day of the week, and a measure of how busy the schedule is at a given moment. By focusing on these time-related factors in our exploratory data analysis and in our machine learning model, we can better predict whether a delay will occur. The paper also helps us understand why certain time periods, such as evenings or peak holiday seasons, tend to have higher delay rates [7].

## 5.5 Paper 5 — Predictive Modeling of Flight Delays at an Airport Using Machine Learning Methods

### Paper Review

This paper studies flight delay prediction using machine learning at a specific airport. The authors used flight data from a Turkish airport collected over three years. They tested several models, including Logistic Regression, Random Forest, XGBoost, CatBoost, and LightGBM, in order to compare their performance. The best result was obtained from XGBoost, with an accuracy of about 80%. The paper also examined whether adding weather data improved the predictions, but the improvement was small in their setting [8].

### Connection to Our Project

This paper is closely related to our project because we are also working on flight delay prediction. Our project focuses on airport punctuality across five European countries: Ireland, Portugal, Belgium, Denmark, and Austria. The paper helps us understand how to use machine learning models to predict whether a flight will be delayed or not, and it shows that comparing more than one model is a good practice. It also gives us realistic expectations about the kind of accuracy we may achieve and reminds us that adding more features (such as weather) does not always improve a model by a large margin [8].

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